

STUDENT ACTIVITY

Workflow Sequencing — Activity

Activity: Flask AI Web App Deployment Pipeline Mapping

Q: Read the project development scenario below. Map out the correct multi-step lifecycle for building an AI-powered web application by ordering the following five workflow stages from first to last:

- Host the application online via a deployment platform.
- Connect the saved Machine Learning model to the Flask backend.
- Gather and prepare the training data.
- Build and train the Machine Learning model.
- Design the frontend user interface using HTML templates.

Next, utilizing the provided text, describe the primary function of routes and templates within the Flask framework in three sentences. Ensure you explicitly mention the Jinja2 Template Engine and how URLs are mapped.

The Scenario: A student team is building a Final Project Demonstration for a Student Performance Predictor. They need to transition their local machine learning project into a fully interactive web application accessible through a browser. The team must sequentially execute all required steps, starting from raw data handling, moving through model creation and backend integration, designing the frontend, and finally making the service publicly accessible on the internet.

Workflow Sequencing — Activity

AI Web Application Development Pipeline

Correct Order of the Workflow Stages

Step	Workflow Stage	Explanation
1	Gather and prepare the training data.	Collect, clean, and preprocess the dataset to ensure it is suitable for training the Machine Learning model.

Step	Workflow Stage	Explanation
2	Build and train the Machine Learning model.	Develop the Machine Learning model using the prepared data and save the trained model for later use.
3	Connect the saved Machine Learning model to the Flask backend.	Load the trained model into the Flask application so it can process user inputs and generate predictions.
4	Design the frontend user interface using HTML templates.	Create user-friendly web pages using HTML and Jinja2 templates for user interaction.
5	Host the application online via a deployment platform.	Deploy the Flask application to a cloud hosting platform so it is accessible through a web browser.

Function of Routes and Templates in Flask

In **Flask**, **routes** map **URLs** to specific Python functions, determining which function is executed when a user visits a particular web address. **Templates** are HTML files rendered using the **Jinja2 Template Engine**, which allows dynamic content to be displayed by combining HTML with Python data. Together, routes handle user requests while Jinja2 templates generate interactive web pages and return them to the user's browser.

Flask AI Web Application Deployment Pipeline

Step	Workflow Stage
1	Gather and prepare the training data.
2	Build and train the Machine Learning model.
3	Connect the saved Machine Learning model to the Flask backend.
4	Design the frontend user interface using HTML templates.
5	Host the application online via a deployment platform.

Routes and Templates in Flask

- **Routes** map **URLs** to Python functions that handle user requests and return responses.
- **Templates** are HTML pages rendered using the **Jinja2 Template Engine**, allowing dynamic content to be displayed.
- Together, routes process incoming requests, while Jinja2 templates generate and display interactive web pages for users.